Anton Matiash

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Professional Summary

Mechanical Engineering undergraduate, ranked 1st in cohort, with research experience in energy-efficient legged robots and a focus on smart, bio-inspired robot bodies and learned control for safe human-robot interaction. Background in competitive physics (IYPT Gold).

Education

BEng in Mechanical Engineering *Sep 2024 – Jun 2027 (expected)*
Swansea University *Swansea, UK*
First Year Average: 89% (Top 1 in cohort), Second Year Average: 86% (Top 1 in cohort)
Selected modules: Data Analysis and Simulation (98%), Engineering Design 1 (95%), Engineering Mathematics (91%), Machines and Control (99%), AI, Statistics and Machine Learning (80%), Engineering Design 2 (81%), Dynamics I (94%)

Modern Robotics specialisation (MR) *Jun 2025 – Oct 2025*
Northwestern University/Coursera *Online*
Completed all 6 courses in specialisation: Foundations of Robot Motion, Robot Kinematics, Dynamics, Motion Planning and Control, Manipulation and Wheeled Mobile Robots, Capstone Project.

High School Diploma *Sep 2017 – Jun 2024*
Kharkiv Physics-Mathematics Science Lyceum №27 *Kharkiv, Ukraine*
Last year Average: 11.1 out of 12. Top 2 in cohort

Experience

Research Intern *Jun 2026 – Sep 2026*
Bristol Robotics Laboratory *Bristol, UK*

- Extended the group's cam-based PEJ concept into an adaptive PEJ model, and trained RL policies in Isaac Lab for a Unitree Go2 and Franka Panda, demonstrating its potential for high energy efficiency.
- Iterated the training pipeline to reduce the expected sim-to-real gap ahead of hardware testing, adding more realistic actuator models and improving the physical imitation learning pipeline.
- Compressed findings into a scientific paper, which is in preparation for submission.

Makerspace Supervisor *Oct 2025 – Aug 2027*
CREU: The Manufactury *Swansea, UK*

- Helped manage a student makerspace, maintained 3D printers, laser cutters, and other equipment. Helped establish basic working procedures for the makerspace and trained new students on how to use the equipment safely.

Secretary *Jul 2026 – Jul 2027*
Swansea University Table Tennis Club *Swansea, UK*

- Ran club administration and organised tournaments of 30+ participants.

Honors & Awards

International Physicists' Tournament 2025 *Kielce, Poland*
4th of 17 countries as part of Team Ukraine: experiments, simulation and theory on Slinky staircase motion; top report score in team.

International Young Physicists' Tournament 2024 (38 countries) *Budapest, Hungary*
Achieved Gold Medal as part of Team Ukraine (2nd of 38); soap-film dynamics problem; top report score in team.

All-Ukrainian Student Physics Olympiad 2023/2024 *Ivano-Frankivsk, Ukraine*
3rd-degree Diploma (national stage, 12th out of ≈ 30 finalists).

Selected Projects

Sprawled vs Conventional Quadruped: Design and RL Comparison *Oct 2026 – Present*
Fusion360, C++, IsaacSim/IsaacLab, Python

- Designing sprawled-leg and conventional quadrupeds of matched mass around Unitree Go actuators, to compare locomotion performance via curriculum RL in Isaac Lab, and preparing a sim-to-real-ready ROS 2 stack.
- Evaluating whether Parallel Elastic Joints reduce energy consumption more for the sprawled design.

Custom Robotics Actuator *Oct 2025 – May 2026*
Fusion360, C++, IsaacSim, Python

- Designed, built and tested a highly modular custom robotics actuator using off-the-shelf BLDC motor and ODrive clone. Compared planetary and harmonic gearboxes and selected 7:1 planetary gearbox to achieve quasi-direct drive performance.

Mobile Manipulator Control for Pick-and-Place (MR Capstone) *Jun 2025 – Oct 2025*
MATLAB, CoppeliaSim [Certificate Link][Demonstration Video]

- Built a full kinematics, trajectory generation and control stack in MATLAB for a KUKA mobile manipulator performing pick-and-place in CoppeliaSim.

Technical Skills

Programming: Python, C++, MATLAB, C#, TypeScript, SQL

Robotics and ML: Isaac Sim/Lab, ROS 2, PyTorch, Simulink

Design and Prototyping: Fusion 360 (CAD/CAM), 3D printing, laser cutting, electronics and soldering

Tools: LaTeX, Git, Docker

Languages: English (fluent), Russian and Ukrainian (native)